ASCII MASTER FLOWCHART

Concepts, Risk, Resilience and Sendai

12-panel revision companion | GS-III Disaster Management

RISK GRAMMAR	MANAGEMENT CYCLE	SENDAI SPINE

DEFINITIONS -> DISTINCTIONS -> POLICY LEVERS -> ANSWER BOUNDARY

ASCII MASTER FLOW - PANEL 1/12: Risk grammar and the disaster threshold

```
HAZARD -> dangerous condition or event with potential to harm
EXPOSURE -> people, livelihoods and assets in its path
VULNERABILITY -> conditions that turn exposure into harm
CAPACITY -> strengths, resources, institutions and measures
INTERACTION -> DISASTER RISK
THRESHOLD -> losses "beyond a community's ability to cope" = DISASTER
RULE -> same hazard magnitude, different outcome, because of E, V and C
```

ASCII MASTER FLOW - PANEL 2/12: Hazard taxonomy by source

```
NATURAL
  GEOPHYSICAL -> earthquake, tsunami, volcanic activity, landslide
  HYDROLOGICAL -> flood, flash flood, debris flow
  METEOROLOGICAL -> cyclone, storm, extreme temperature
  CLIMATOLOGICAL -> drought, desertification, forest fire
  BIOLOGICAL -> epidemic, pest infestation
SOCIO-NATURAL -> natural process amplified by human decisions
ANTHROPOGENIC -> pollution, deforestation, chemical spillage, accidents
USE -> family choice fixes the dominant mitigation instrument
```

ASCII MASTER FLOW - PANEL 3/12: Onset speed and creeping emergencies

```
RAPID ONSET -> earthquake, cyclone, flood, tsunami
  definable impact moment -> warning, evacuation, rescue, relief
SLOW ONSET / CREEPING EMERGENCIES -> climate change, desertification,
  soil degradation, drought
  no impact moment -> progressive loss of the ability to derive sustenance
  requirement -> must find adequate reflection in preparedness planning
GAP -> institutions built for acute events under-serve the slow category
```

ASCII MASTER FLOW - PANEL 4/12: The five-fold vulnerability typology

```
ECONOMIC -> income diversity, access to means of production
PHYSICAL -> proximity to hazard, construction type
SOCIAL -> poor, pregnant and lactating women, disabled, children, elderly
ENVIRONMENTAL -> ecosystem sensitivity, lost natural buffers
ATTITUDINAL -> negative disposition to change, dependence on outside help
RULE -> coinciding dimensions compound harm, they do not merely add
MARK -> the attitudinal dimension is the one most answers omit
```

ASCII MASTER FLOW - PANEL 5/12: Risk definition and the four treatment options

```
RISK = expected losses | GIVEN AREA | SPECIFIC TIME PERIOD
WORKING RELATION = PROBABILITY OF HAZARD x DEGREE OF VULNERABILITY
USE -> order the problem and locate the movable factor
DO NOT USE -> arithmetic prediction of the next event
TREATMENT
  ACCEPTANCE -> informed decision to bear the consequence
  AVOIDANCE -> do not undertake the risk-generating activity
  REDUCTION -> lower the likelihood or the consequence
  TRANSFER -> shift the burden, for example insurance
LIMIT -> only avoidance and reduction change the loss; residual risk remains
```

ASCII MASTER FLOW - PANEL 6/12: Two-phase cycle and the 3Rs to 3Ps shift

```
PRE-DISASTER: RISK MANAGEMENT PHASE
  PREVENTION -> MITIGATION -> PREPAREDNESS
IMPACT
POST-DISASTER: CRISIS MANAGEMENT PHASE
  RELIEF -> RESPONSE -> REHABILITATION -> RECONSTRUCTION -> RECOVERY
APPROACH SHIFT
  OLD 3 Rs -> RESCUE, RELIEF, RESTORATION (reactive)
  NEW 3 Ps -> PREVENTION, PREPAREDNESS, PROOFING (anticipatory)
POINT -> restoration reinstates the failed condition; proofing does not
```

ASCII MASTER FLOW - PANEL 7/12: Pre-impact instruments

```
PREVENTION -> can the risk be avoided? relocation, prohibition in floodway
MITIGATION -> can the impact be lessened?
  STRUCTURAL -> embankments, retrofitting, cyclone shelters
  NON-STRUCTURAL -> land-use zoning, building codes, awareness, incentives
PREPAREDNESS -> can we act in time? plans, stocks, warning, drills, teams
SEQUENCE -> the three are complementary, never substitutable
TRAP -> relocation is prevention, not mitigation
```

ASCII MASTER FLOW - PANEL 8/12: Post-impact ladder and the recovery fork

```
RELIEF AND RESPONSE -> life safety, medical care, shelter, water
REHABILITATION -> services, schooling, health access, livelihoods
RECONSTRUCTION -> housing, infrastructure, productive assets
RECOVERY -> process of restoring or improving functioning; sets the next risk profile
FORK
  RESTORE AS BEFORE -> the same vulnerability returns
  BUILD BACK BETTER -> safer siting, stronger codes, restored buffers
RISK-CREATION LOOP -> development choice raises exposure and vulnerability
  even where hazard frequency is unchanged; recovery recreates or reduces it
```

ASCII MASTER FLOW - PANEL 9/12: DRR definition and field of action

```
DRR = reduced EXPOSURE + lessened VULNERABILITY of people and property
      + wise management of LAND and ENVIRONMENT + improved PREPAREDNESS
SCOPE -> prevent new risk, reduce existing risk, manage residual risk
FIELD OF ACTION (sequence with feedback)
  POLICY AND LEGAL FRAMEWORK -> RISK ASSESSMENT -> RISK AWARENESS
  -> PLAN IMPLEMENTATION -> EARLY WARNING SYSTEMS -> USE OF KNOWLEDGE
DIAGNOSTIC -> name the broken stage; everything downstream consumes
  the output of risk assessment
```

ASCII MASTER FLOW - PANEL 10/12: Resilience, elements and measurement limit

```
RESILIENCE VERBS -> RESIST, ABSORB, ACCOMMODATE, ADAPT, TRANSFORM, RECOVER
  in a timely and efficient manner, preserving and restoring essential
  basic structures and functions
DETERMINED BY -> hazard, exposure, vulnerability and capacity configuration
FRAMEWORK ELEMENTS -> RISK KNOWLEDGE, ROBUSTNESS, REDUNDANCY,
  RESOURCEFULNESS, RAPID RECOVERY, ADAPTIVE CAPACITY
TRAP -> "bouncing back" restores the condition that failed
LIMIT -> measurement needs capacity data and a stated metric
```

ASCII MASTER FLOW - PANEL 11/12: India's instruments and the NDMP

```
BEFORE SENDAI
  DISASTER MANAGEMENT ACT, 2005 -> statutory architecture
  NATIONAL POLICY ON DISASTER MANAGEMENT, 2009 -> prevention orientation
AFTER SENDAI
  NDMP 2016 -> five Thematic Areas for Action
    UNDERSTANDING RISK | INTER-AGENCY COORDINATION
    INVESTING IN DRR (STRUCTURAL) | INVESTING IN DRR (NON-STRUCTURAL)
    CAPACITY DEVELOPMENT
  PM TEN-POINT AGENDA -> AMCDRR, NEW DELHI, NOVEMBER 2016,
    each point aligned to Sendai priorities of action
  NDMP 2019 -> revised and latest published edition; cite as operative
CRITIQUE OF 2016 EDITION -> too generic, no time frame, no fund projection,
  no goals or targets of its own
```

ASCII MASTER FLOW - PANEL 12/12: Hyogo to Sendai, Targets A-G and the answer boundary

```
HYOGO 2005-2015 -> five priorities, no quantified targets, voluntary
SENDAI 2015-2030 -> adopted 18 MARCH 2015, Third UN World Conference,
  Sendai, Japan; voluntary and non-binding; four priorities
  1 UNDERSTAND RISK | 2 STRENGTHEN RISK GOVERNANCE
  3 INVEST IN DRR FOR RESILIENCE | 4 PREPARE AND BUILD BACK BETTER
TARGETS
  A mortality | B affected people | C economic loss / global GDP
  D critical infrastructure and basic services -> loss outcomes, 2030
  E national and local DRR strategies by 2020 -> only 2020 deadline
  F international cooperation | G multi-hazard early warning and risk
  information -> means of implementation
CURRENT ANCHORS -> EW4All (call March 2022, COP27 November 2022, by 2027;
  UNDRR, WMO, ITU, IFRC pillars) | MIDTERM REVIEW 17-19 MAY 2023,
  Political Declaration as UNGA RESOLUTION 77/289, 18 MAY 2023
BOUNDARY -> frameworks, strategies and initiatives are inputs;
  avoided loss and coverage require separate dated outcome evidence
```